Control Systems I 2004 / Final examination answer sheet

Student ID Number Name

NOTE: You can use both sides of this sheet. Clearly indicate the question number so that the grader can identify where the answer is written.

Q.1 (1) The definition of equilibrium is the states that remain unchanged. This means that all time derivatives in the equation of motion vanish. From (b), this gives $mgl\sin\theta = 0$ so $\theta = 0, \pi$ and any value of q.

(2) Nonlinear terms are any terms that are not constants or first order polynomial. From (a), $ml\cos\theta\,\ddot{\theta}, -ml\sin\theta\,\dot{\theta}^2$ are nonlinear terms and so from (b), $-ml\cos\theta\,\ddot{q}$, $mgl\sin\theta$.

(3) State vector $\boldsymbol{x} = \left[q, \theta, \dot{q}, \dot{\theta}\right]^T$

Around $\theta = 0$, the nonlinear terms can be approximated in the following way:

$$ml\cos\theta\,\ddot{\theta} \to ml\ddot{\theta}, -ml\sin\theta\,\dot{\theta}^2 \to 0,\ -ml\cos\theta\,\ddot{q} \to -ml\ddot{q},\ mgl\sin\theta \to mgl\theta$$

Then, $\begin{bmatrix} M+m & -ml \\ -ml & ml^2 \end{bmatrix}\begin{bmatrix} \ddot{q} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F \\ mgl\theta \end{bmatrix} \to \begin{bmatrix} \ddot{q} \\ \ddot{\theta} \end{bmatrix} = \frac{1}{Mml^2}\begin{bmatrix} ml^2 & ml \\ ml & M+m \end{bmatrix}\begin{bmatrix} F \\ mgl\theta \end{bmatrix} = \begin{bmatrix} \frac{mg}{M} \\ \frac{(M+m)g}{Ml} \end{bmatrix}\theta + \begin{bmatrix} \frac{1}{M} \\ \frac{1}{Ml} \end{bmatrix}F$

State equation: $\frac{d}{dt}\boldsymbol{x} = A\boldsymbol{x} + B\boldsymbol{u}$, where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mg}{M} & 0 & 0 \\ 0 & \frac{(M+m)g}{Ml} & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M} \\ \frac{1}{M} \end{bmatrix}$$

Around $\theta = \pi$, the nonlinear terms can be approximated in the following way:

$$ml\cos\theta\,\ddot{\theta} \to -ml\ddot{\theta}, \quad -ml\sin\theta\,\dot{\theta}^2 \to 0, \quad -ml\cos\theta\,\ddot{q} \to ml\ddot{q}, \quad mgl\sin\theta \to -mgl\theta$$

Then $\begin{bmatrix} M+m & ml \\ ml & ml^2 \end{bmatrix}\begin{bmatrix} \ddot{q} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F \\ -mgl\theta \end{bmatrix} \to \begin{bmatrix} \ddot{q} \\ \ddot{\theta} \end{bmatrix} = \frac{1}{Mml^2}\begin{bmatrix} ml^2 & -ml \\ -ml & M+m \end{bmatrix}\begin{bmatrix} F \\ -mgl\theta \end{bmatrix} = -\begin{bmatrix} \frac{mg}{M} \\ -\frac{(M+m)g}{Ml} \end{bmatrix}\theta + \begin{bmatrix} \frac{1}{M} \\ -\frac{1}{Ml} \end{bmatrix}F$

State equation: $\frac{d}{dt}\boldsymbol{x} = A\boldsymbol{x} + B\boldsymbol{u}$, where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mg}{M} & 0 & 0 \\ 0 & -\frac{(M+m)g}{Ml} & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M} \\ -\frac{1}{M} \end{bmatrix}$$

(4) Around $\theta = 0$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mg}{M} & 0 & 0 \\ 0 & \frac{(M+m)g}{Ml} & 0 & 0 \end{bmatrix} \to |\lambda I - A| = \lambda^2 \left(\lambda - \frac{\sqrt{(M+m)g}}{\sqrt{Ml}}\right)\left(\lambda + \frac{\sqrt{(M+m)g}}{\sqrt{Ml}}\right) \to Re\left(\frac{\sqrt{(M+m)g}}{\sqrt{Ml}}\right) > 0 \to \text{unstable}$$

Around $\theta = \pi$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mg}{M} & 0 & 0 \\ 0 & -\frac{(M+m)g}{Ml} & 0 & 0 \end{bmatrix} \to |\lambda I - A| = \lambda^2 \left(\lambda - \frac{i\sqrt{(M+m)g}}{\sqrt{Ml}}\right)\left(\lambda + \frac{i\sqrt{(M+m)g}}{\sqrt{Ml}}\right) \to Re(\lambda) = 0 \to \text{unstable}$$

(Note: By definition, a system is unstable if not all eigenvalues have negative real part. If there existed a friction term, then the latter case must be stable)

Q2. (1) Controllability: Rank $[B, AB, \dots, A^{n-1}B] = n$, Observability: Rank $\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = n$

(2) In the controllable canonical form, the transition matrix has its eigenvalues on the first row. The closed loop with state feedback $u = -Kx$ has the transition matrix in the form as:

$$A = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & & & 0 \\ 0 & 1 & 0 & \ddots & \vdots \\ & 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix} \to A - BK = \begin{bmatrix} -a_1 - k_1 & -a_2 - k_2 & -a_3 - k_3 & \cdots & -a_n - k_n \\ 1 & 0 & & & 0 \\ 0 & 1 & 0 & \ddots & \vdots \\ & 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix}$$

This means that the eigenvalues of the matrix can have arbitrary value by choosing k_is in response to a_is.

(3) Controllability: Rank $[B, AB, \dots, A^{n-1}B] =$ Rank $\begin{bmatrix} 1 & -1 & -1 \\ -1 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} = 2 \to$uncontrollable ($x_3$ is uncontrollable, but stable since $\dot{x}_3 = -x_3$)

Observability when $C = [1\ 0\ 0] \to Rank \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = 3 \to$ observable→ stabilizable

Observability when $C = [0\ 1\ 0] \to Rank \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} = 1 \to x_1, x_3$ are unobservable, so not stabilizable

Observability when $C = [0\ 0\ 1] \to Rank \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} = 1 \to x_1, x_2$ are unobservable, so not stabilizable